

Topic to discuss

- Stirling central difference Interpolation
- Stirling difference table
- Stirling difference Formula
- Numerical Problem
- Homework Problem with solution.

Stirling central difference Interpolation

It is especially useful when x is near the mid-point between two tabulated values.

The formula is more accurate than Gauss Forward/Backward when the chosen origin is at the centre.

Stirling averages forward and backward difference: $\frac{\Delta^n + \nabla^n}{2}$

When to use :

- when x is exactly the middle of the table
- when , $-0.25 \leq u \leq 0.25$

$$\text{where, } u = \frac{x - x_0}{h}$$

Stirling central difference Table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	y_{-2}	$\Delta y_{-2} = y_{-1} - y_{-2}$	$\Delta^2 y_{-2} = \Delta y_{-1} - \Delta y_{-2}$	$\Delta^3 y_{-2} = \Delta^2 y_{-1} - \Delta^2 y_{-2}$	$\Delta^4 y_{-2} = \Delta^3 y_{-1} - \Delta^3 y_{-2}$
x_{-1}	y_{-1}	$\Delta y_{-1} = y_0 - y_{-1}$	$\Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}$	$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}$	
x_0	y_0	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$		
x_1	y_1	$\Delta y_1 = y_2 - y_1$			
x_2	y_2				

Stirling central difference formula

We know,

Gauss Forward difference formula :

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_{-2} + \dots$$

and Gauss Backward difference formula :

$$y(x) = y_0 + u \Delta y_{-1} + \frac{u(u+1)}{2!} \Delta^2 y_{-1} + \frac{(u-1)u(u+1)}{3!} \Delta^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!} \Delta^4 y_{-2} + \dots$$

Taking the mean of Gauss forward and backward formula, we obtain :

$$\begin{aligned} \frac{2y(x)}{2} &= \frac{2y_0}{2} + \frac{u}{2} (\Delta y_0 + \Delta y_{-1}) + \frac{u \Delta^2 y_{-1}}{2!} \left(\frac{u-1+u+1}{2} \right) + \frac{(u+1)u(u-1)}{3! \times 2} (\Delta^3 y_{-1} + \Delta^3 y_{-2}) \\ y(x) &= y_0 + \frac{u}{2} (\Delta y_0 + \Delta y_{-1}) + \frac{u^2}{2!} (\Delta^2 y_{-1}) + \frac{u(u^2-1)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) \end{aligned}$$

Stirling central difference formula

Taking the mean of Gauss forward and backward formula, we obtain:

$$y(x) = y_0 + u \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2-1)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{u^2(u^2-1)}{4!} \Delta^4 y_{-2} + \frac{u(u^2-1)(u^2-2)}{5!} \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right)$$

where, $u = \frac{x - x_0}{h}$

and h is constant difference between successive values of x .

Also, $-0.25 \leq u \leq 0.25$

Numerical Problem :

find the value of y at $x=0.5$, using stirling central difference formula,

x	-2	-1	0	1	2
y	5	2	1	2	5

Solution : step length $= -1 - (-2) = 1$

choose central value closest to $x(0.5)$; $x_0 = 0$

Interpolation point : $x = 0.5$

$$u = \frac{x - x_0}{h} = \frac{0.5 - 0}{1}$$

$$u = 0.5$$

stirling central difference table.

	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	-2	5	-3	2		
x_{-1}	-1	2	-1	2	0	
x_0	0	1	1	2	0	0
x_1	1	2	3	2		
x_2	2	5				

Stirling central difference formula.

$$y(x) = y_0 + u \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2-1)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{u^2(u^2-1)}{4!} \Delta^4 y_{-2}$$
$$= 1 + 0.5 \left(\frac{1 + (-1)}{2} \right) + \frac{0.5^2}{2!} \times 2 + \frac{0.5(0.5^2-1)}{3!} (0) + \frac{0.5^2(0.5^2-1)}{4!} \times 0$$

$$= 1 + 0 + 0.25$$

$$= 1.25$$

Ans.

Homework Problem

Using Stirling formula, find the value of y at $x = 2.5$

x	0	1	2	3	4
y	3	6	11	18	27

Solution : Step length, $h = 1 - 0 = 1$

Choose central value closest to $x(2.5)$; $x_0 = 2$

Interpolation point, $x = 2.5$

$$\therefore u = \frac{x - x_0}{h} = \frac{2.5 - 2}{1} = 0.5$$

$$u = 0.5$$

Stirling central difference table for given data points,

	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	0	3	3	2		
x_{-1}	1	6	5	2	0	
x_0	2	11	7	2	0	0
x_1	3	18	9	2		
x_2	4	27				

Applying stirling central difference formula,

$$y(x) = y_0 + u \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2-1)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{u^2(u^2-1)}{4!} \Delta^4 y_{-2}$$

$$= 11 + 0.5 \left(\frac{7+5}{2} \right) + \frac{0.5^2}{2!} \times 2 + 0 + 0$$

$$= 11 + 3 + 0.25$$

$$= 14.25 \text{ Ans.}$$

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